Analytics

Formulas & Definitions

This page explains all calculations and metrics used in the Customer Analytics and Coupon Analytics dashboards.

Key Metric Definitions

Period Metrics

New Members (Period)
Customers whose Registration Date falls within the selected period

Returning Customers (Period)
Count of customers with at least 1 return visit in the period

Active (Period)
Total unique customers (VIP IDs, excluding VIP ID = 0) who made purchases in the period

Return Visit Rate (Period)
(Returning Customers / Active Customers) × 100

Invoices (Customers)
Total invoices from customers with valid VIP IDs (excludes VIP ID = 0)

Amount (Customers)
Sum of sales_amount for transactions with valid VIP IDs (excludes VIP ID = 0)

Total Invoices
All invoices in period including VIP ID = 0

Total Amount
Sum of all sales_amount in period including VIP ID = 0

All-Time Metrics

Total Customers (All Time)
Total count of records in Customer table

Member Active (All Time)
Customers with points > 0

Member Inactive (All Time)
Customers with points = 0

Return Visit Rate (All Time)
(Returning Customers / Total Customers in DB) × 100

Customer Analytics Formulas

1. Return Visits Calculation ⚠ CRITICAL FORMULA

The **Return Visits** metric counts how many invoices a customer had AFTER their first visit.

⚠ **IMPORTANT:** This formula considers the **Registration Date** to determine if the first purchase is a "first visit" or a "return visit".

```
Complete Formula:

IF first_purchase_date == registration_date:
    return_visits = total_purchases - 1
    # First invoice on reg day = first visit (NOT a return)
    # All subsequent invoices = return visits

ELSE (first_purchase_date != registration_date OR no reg date):
    return_visits = total_purchases
    # Customer already visited to register
    # ALL purchases = return visits

is_returning = (return_visits > 0)
Note: We count INVOICES, not unique days
```

Key Insight: The formula is applied **CONSISTENTLY** at all levels (Global, Shop, Season), but uses **DIFFERENT purchase lists** for each context, resulting in **different "first_purchase_date"** values.

Example 1: Registration day purchase (Same Day)
- Registration Date: **2026-01-01**
- Purchases: **2026-01-01** (3 invoices on same day)
- first_date == reg_date ✅
- **Return Visits = 3 - 1 = 2**
- Is Returning = True (had 2 return visits on reg day)

Example 2: Registered before, buying later (Different Day)
- Registration Date: **2026-01-01**
- Purchases: **2026-02-15** (2 invoices)
- first_date != reg_date ✅
- **Return Visits = 2** (all purchases are returns)
- Is Returning = True

Example 3: Single purchase on registration day
- Registration Date: **2026-01-01**
- Purchases: **2026-01-01** (1 invoice)
- first_date == reg_date ✅
- **Return Visits = 1 - 1 = 0**
- Is Returning = False (no returns yet)

Example 4: No registration date
- Registration Date: **NULL**
- Purchases: **2026-03-10** (2 invoices)
- first_date != reg_date ✅ (NULL comparison)
- **Return Visits = 2** (assume already visited to get VIP ID)
- Is Returning = True

🔍 Context-Dependent Calculations

CRITICAL: The same customer can have DIFFERENT returning status in different contexts!

⚠️ **Why Totals Don't Match:** This is **CORRECT** behavior, not a bug!
Global returning invoices ≠ Sum of shop returning invoices ≠ Sum of season returning invoices

Real Example - Customer A:

Registration: 2026-01-01
Purchases:
• Invoice #1: Shop X, 2026-01-01 (reg day)
• Invoice #2: Shop Y, 2026-02-01

• Invoice #3: Shop C, 2026-03-01
• Invoice #4: Shop C, 2026-03-01

CONTEXT	PURCHASES	FIRST DATE	REG DATE	FORMULA	RETURN VISITS	IS RETURNING?	RET INVOICES
GLOBAL	4 invoices	1/1	1/1	4 - 1 = 3	3	YES	**4**
Shop X	1 invoice	1/1	1/1	1 - 1 = 0	0	NO	0
Shop Y	1 invoice	2/1	1/1	1 (!=)	1	YES	1
Shop C	2 invoices	3/1	1/1	2 (!=)	2	YES	2
TOTALS:							
Global Returning Invoices:							**4**
Sum of Shop Returning Invoices (X+Y+C):							**0+1+2 = 3**
Difference:							**1 invoice**

Why Different?
• **Shop X:** First purchase at Shop X on reg day (1/1 == 1/1) → NOT returning at Shop X
• **Shop Y:** First purchase at Shop Y after reg day (2/1 != 1/1) → IS returning at Shop Y
• **Globally:** First purchase on reg day (1/1 == 1/1), but has 4 total → IS returning globally

✅ **This is CORRECT!** Same customer, different "first_purchase_date" per context.

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2. Return Visit Rate

Return Visit Rate (Period) % = (Returning Customers / Active Customers) × 100

- **Returning Customers**: Number of customers with return_visits > 0
- **Active Customers**: Total unique customers who made purchases in the period

Example: In January 2026:
- 100 unique customers made purchases
- 75 of them have return_visits > 0
- **Return Visit Rate = (75 / 100) × 100 = 75%**

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3. VIP ID = 0 (Buyer Without Info)

Purchases with `VIP ID = 0`, blank, or null are **EXCLUDED** from customer metrics but **INCLUDED** in total invoice/amount metrics.

Invoices (Customers)
Total invoices from customers with valid VIP IDs (excludes VIP ID = 0)

Amount (Customers)
Sum of sales_amount from customers with valid VIP IDs (excludes VIP ID = 0)

Total Invoices
ALL invoices including VIP ID = 0

Total Amount
Sum of ALL sales_amount including VIP ID = 0

> **Example:**
> - 100 customers made 500 invoices totaling 250M VND
> - 50 invoices (25M VND) have VIP ID = 0
>
> **Results:**
> - Active (Period): 100 (excludes VIP 0)
> - Invoices (Customers): 450
> - Amount (Customers): 225M VND
> - Total Invoices: 500 (includes VIP 0)
> - Total Amount: 250M VND (includes VIP 0)

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4. Season (Year-Aware)

Seasons are defined as:

- **M2-4** (Feb–Apr): February, March, April
- **M5-7** (May–Jul): May, June, July
- **M8-10** (Aug–Oct): August, September, October
- **M11-1** (Nov–Jan): November, December, January (spans 2 years)

Each season is labeled with the year(s) it covers:

> **Examples of year-aware season labels:**
> - `M2-4 2025`: Feb-Apr 2025

- `M11-1 2025-2026`: Nov 2025 + Dec 2025 + Jan 2026
- `M11-1 2024-2025`: Nov 2024 + Dec 2024 + Jan 2025

For date range Jan 2025 to May 2026, the "By Season" table shows all season windows with data:

- M11-1 2024-2025 (if Jan 2025 has data)
- M2-4 2025
- M5-7 2025
- M8-10 2025
- M11-1 2025-2026
- M2-4 2026
- M5-7 2026 (if May 2026 has data)

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5. VIP Grade Order

Grades are always sorted in this order:

1. No Grade
2. Member
3. Silver
4. Gold
5. Diamond

Note: "Olden" and "Golden" are automatically normalized to "Gold"

🎫 Coupon Analytics Formulas

1. Coupon Usage Percentages

By Shop Table shows two percentage calculations:

```
% Used / Total (Period):
= (Used Coupons at Shop / Total Coupons in Period - All Shops) × 100

% Used / Total Used (All Shops):
= (Used Coupons at Shop / Total USED Coupons - All Shops) × 100
```

Example: Period has 1000 coupons total across all shops, 400 are used.
- Shop A used 100 coupons
- **% Used / Total (Period) = (100 / 1000) × 100 = 10%**
- **% Used / Total Used (All Shops) = (100 / 400) × 100 = 25%**

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2. Coupon ID Prefix Filter

The prefix filter is **case-insensitive** and matches the beginning of the Coupon ID.

Example: Filter = `SEMIR50K`
Matches:
- SEMIR50K001
- SEMIR50K002
- semir50k003 (case-insensitive)

Does NOT match:
- ABC-SEMIR50K (prefix must be at start)
- SEMIR40K001 (different prefix)

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3. Customer Name & Phone in Coupon Detail

When a coupon is used, we lookup the customer's name and phone number using the VIP ID from the matched invoice.

- If no invoice match: VIP ID, Name, Phone are blank
- If invoice found but VIP ID = 0: Name = "-", Phone = "-"
- If invoice found with valid VIP ID: Name and Phone from Customer table